Competent

29/30

97%

TOOLS, EQUIPMENT AND MATERIALS

FORMATIVE ASSESSMENT

Module #	671101000-KM-02
NQF Level	level 4
Notional hours	80
Credit(s)	Cr 8
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

Details	Please Complete this Section
Name	SELA PHIRI
Contact Address	47 4724 EVERLASTING STREET

Telephone (H)	
Telephone (W)	
Facsimile	NA
Cellular	062 560 7912
E-mail	Sthaphiri12@gmail.com

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About the Formative Assessment	
Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment

	criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room.
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ **points.** In order to meet the pass mark, learners are required to achieve a minimum of **80%** of the total marks.

LEARNER ASSESSMENT PLAN

code	/MODULE TITLES	NQF level	Credits	✓

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Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
	Assessment contract & assessment plan	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1st Reassessment	Assessors summary report / instruments	Learner file Date:	
2nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place	Eastc Electrical
Language	English
Resources	
Barriers	N/A

5

People to be involved with assessment

Learner: Sela Phiri	Manager:
Trainer: Miss R - Nyakala	Mentor / Coach:
Facilitator Miss R - Nyakala	
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Learners Name: Sela Phiri

Learner's signature: _____ Date: _____

Facilitator Name: Ruth Nyakala

Facilitator's signature: _____ Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
   - If the laid down assessment procedures were not followed during assessments
   - If not all evidence available was taken into account during the assessment

- The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
- The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
- Not all the range items were available for assessment

2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.

3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal "form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.

4. Should the internal moderator re-affirm the assessor' decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

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A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

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Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: S. _____ Date: _____

Employee witness: _____ Date: _____

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PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate **and put them at ease**	✓
2	**Explain the purpose of the meeting** (why you are there and how long the meeting will take)	✓
3	**Explain the** ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	✓
4	**Explain** ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	✓
5	**Explain the assessment process?** ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	✓
6	**Give Learner copies of the following documentation and explain each document** ▪ The Assessment Guide which includes o The relevant unit standard (s) o Assessment contract o Assessment plan o Observation checklist o Knowledge checklist	✓
7	**Discuss the assessment plan** (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	✓
8	**Tell the candidate his/her rights and responsibilities, the assessment procedures and policies** ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	✓
9	**Ensure the assessment environment is appropriate** or make special arrangements	✓
10	Discuss moderation	✓
11	Allow the learner opportunity to clarify any items discussed	✓

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: Sela Phiri	Signature S. Phiri
Facilitator Name:	Signature:

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Complete the following questions as per instructions provided

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LEARNING UNIT 1:

KM-02-KT01: Hand tools and power tools (50%)

Learning Outcome 1:	KT0101 Types, uses and care of hand tools
Learning Outcome 2:	KT0102 Types, uses and care of portable power tools
Learning Outcome 3:	KT0103 Types, uses and care of fixed power tools

Task	Questions Description	Marks
1.	Identify portable power tools and describe their uses Identify fixed power tools and describe their uses	10

Portable power tool	Use
Electric Drill	used to drill holes in wood, metal, plastic or walls and to drive screws
Angle Grinder	for cutting, grinding and polishing metal or stone
JigSaw	to cut curved or irregular shape in wood plastic or metal
Heat Gun	to shrink electrical insulation, remove paint or soften materials
Impact Drill	drilling into hard material such as concrete or brick

Fixed power tools and their uses

Fixed power tools	Use
Bench Grinder	used for sharpening tools, grinding metal and removing rough edges
Band saw	to cut curves and irregular shape in wood or metal

Table Saw	to cut large pieces of wood accurately in straight lines
lathe Machine	for shaping metal or wood by rotating the material.
Drill press	Precise drilling holes

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LEARNING UNIT 2:

KM-02-KT02: Measuring and testing instruments (50%)

Learning Outcome 1:	KT0201 Portable electrical measuring instruments
Learning Outcome 2:	KT0202 Types of electrical measuring and testing instruments
Learning Outcome 3:	KT0203 Safe use of measuring and testing instruments
Learning Outcome 4:	KT0204 Construction and operating principles of measuring and testing instruments
Learning Outcome 5:	KT0205 Methods of connecting measuring and testing instruments in circuits
Learning Outcome 6:	KT0206 Applications and methods of using electrical measuring and testing instruments
Learning Outcome 7:	KT0207 Care and maintenance of single and three phase measuring and testing instruments
Learning Outcome 8:	KT0208 Instrument transformers
Learning Outcome 9:	KT0209 Types and functions of panel mounted electrical measuring and testing instruments
Learning Outcome 10:	KT0210 Safe use of panel mounted measuring and test instruments
Learning Outcome 11:	KT0211 Protection of measuring instruments when connected in a circuit.

Task	Questions Description	Marks
2	a) Identify different types of electrical measuring instruments including fixed and portable and state the purpose. b) Explain by means of circuit diagrams how a ammeter, voltmeter is directly connected and	5

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2. Portable Electrical measuring instruments

instrument	Purpose
Ammeter	measures electric current (A) in a circuit
Voltmeter	measures voltage (V)
Ohmmeter	measures resistance (Ω) of component, wire or circuit
Multimeter	measuring voltage, current and resistance

Fixed electrical measuring

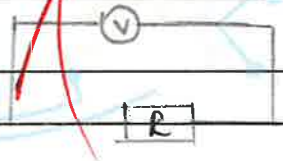
instrument	purpose
Panel Ammeter	Continuously measures current in a fixed system
Panel voltmeter	monitors voltage in power distribution panels
Power meter	measures power consumption (kW, kVA, kWh) in a fixed system
Frequency meter	measures the frequency of AC supply in Hz

2. (b) direct Connection of an ammeter

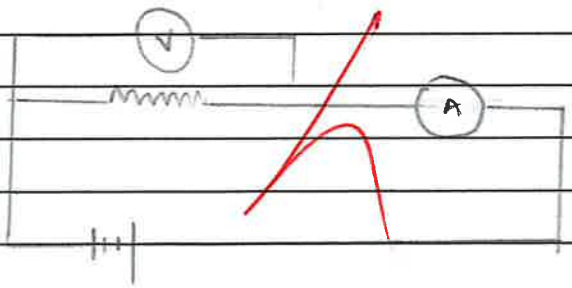


the ammeter is placed in series with resistor

Direct connection of voltmeter



The voltmeter is connected across the resistor



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	indirectly connected by using instrument transformers.	

instrument transformers act as the bridge between the dangerous high-power side and the low-power side where your testing instrument live.

Current Transformers (CTs) - are connected in series with the line. They "step down" the high current to a measurable level for your ammeter

- Pontential transformers (PT's / VTS : are connected in parralel. They step down the high system voltag to a standard level (usually 110V) so that your volt meter can display the value safety

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Task	Questions Description	Marks
3	Explain the function of the wattmeter (electrodynamometer type), kilowatt-hour meter, frequency meter, power factor meter and maximum demand meter	5

1. Wattmeter (electrodynamometer) - measures active power (watts W) in a circuit.
- Can be used in AC or DC circuits

2. kilowatt-hour (kWh) meter
measures total energy consumed over time in kilowatt hours

3. Frequency meter - measures the frequency of an AC supply in Hertz (Hz)

4. Power factor meter - measures the power factor ($\cos\theta$) of an AC circuit

5. Maximum Demand meter - measures the highest load (kW or kVA) drawn over a specified period typically 15-30 minutes

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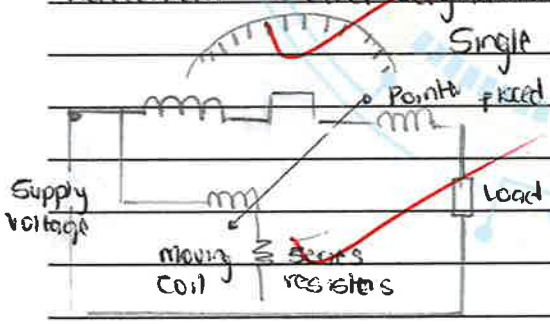
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Task	Questions Description	Marks
4	Illustrate, by means of circuit diagrams, how the wattmeter (electrodynamometer type), kilowatt-hour meter, frequency meter, power factor meter and maximum demand meter are connected in single and three phase circuits:	10

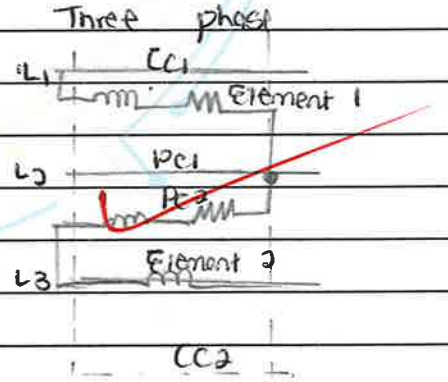
Instrument	Single phase	Three phase
wattmeter	Series CC parallel VC	Two - watt meter method
kWh meter	Series CC, parallel VC	3 CCs and 3 VCs
Frequence meter	Across phase and Neutral	Between any two phases
power factor meter	Series CC ~~Parrel~~ Parallel VC	Two - wattmer method
maximum demand meter	Series CC, parallel VC	Through CT's and PTs

wattmeter (electrodynamometer)

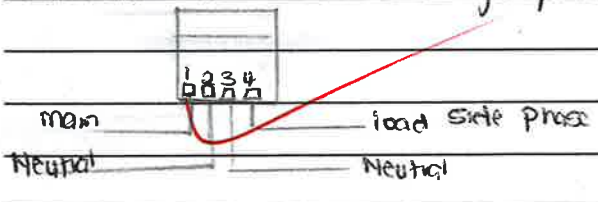
Single phase



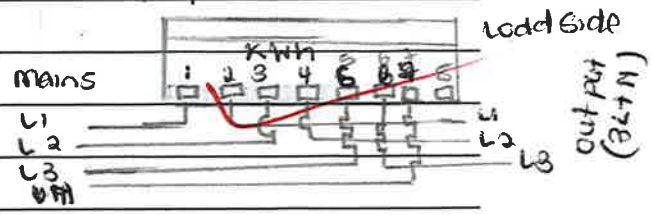
Three phase



kilowatt-hour meter (single phase



Three phase

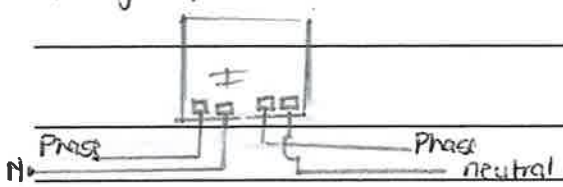


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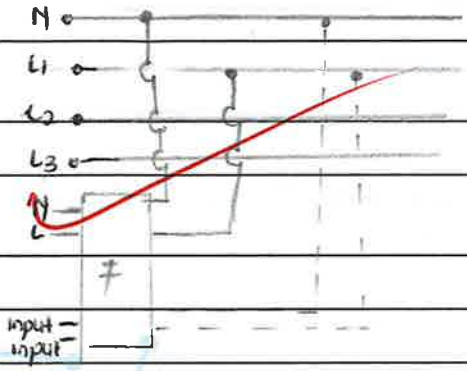
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Frequency meter

Single phase

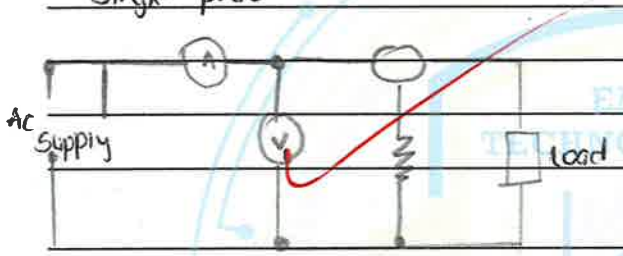


Three phase

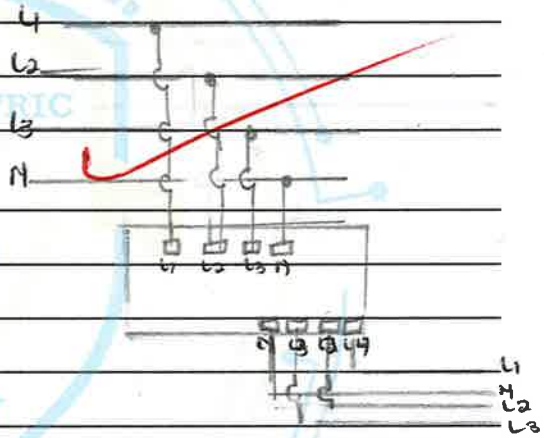


Power Factor meter

Single phase

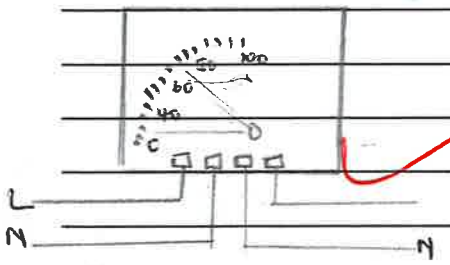


Three phase

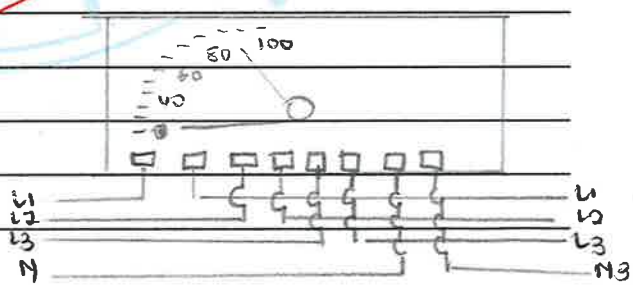


Maximum demand meter

Single phase



Three phase



SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

OUTCOMES		1	2	3
1	Portable electrical measuring instrument			✓
2	types of electrical measuring and testing instru			✓
3	Safe use of measuring and testing instru			✓
4	instrument transformers			✓

S. [redacted]

Learner Signature **Date**

Facilitators' Signature **Date**

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ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS
Well done

ASSESSMENT JUDGEMENT
Learner's Total Mark: 97% Requirements met ☒ Requirements not met ☐
Action/s required: N/A By when:

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR
I RUTH NYAKALA (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

5

DECLARATION BY LEARNER			
I, ... Sela Phiri .....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
I ..... Ruth Nkala ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	**Date**	**Moderator**	**Date**

5

Application for Assessment

First Attempt ☐ Second Attempt ☐ Third Attempt☐ RPL ☐

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

Notification of Assessment

Assessment will take place as follows:

	Theory	**Practical**
Date		
Time	9H00	9H00
Venue	Employer premises	Employer premises
Instruments	Written assignment	Practical Assessment

Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

ETDP PRACTITIONER

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Assessment Planning

	Theory	Practical
Date of assessment:		
When would be the best time for the Assessment?	9H00	9H00

Have you received an application to assess the candidate? Yes
What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method. 1. Written Assignment : Printed Questionnaire 2. Observation : As per Practical Assessment Guide 3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation
What resources would you require to conduct the assessment? 1. Assessor toolkit 2. Theoretical assignment 3. Practical assessment 4. Learner toolkit
How would you sequence your activities? Explain 1. Record preparation process 2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment 3. Learner will write summative assessments

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4. Conduct observation in workplace where applicable
5. Complete learner toolkit
6. Complete assessor toolkit

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